

Vulnerability Scanner (Mini Project)

Project Overview

Build a simple script to detect common vulnerabilities in a web application or network.

Key Features

- Scan for open ports or weak configurations
- Identify outdated software versions
- Generate a simple vulnerability report

Technologies Used

Python, Socket Programming, Requests Library, JSON Report Generation

Folder Structure

```
vulnerability-scanner/  
├──  
│   ├── scanner.py  
│   ├── report.json  
│   ├── requirements.txt  
│   └── README.md
```

Install Required Packages

```
pip install requests python-nmap
```

scanner.py

```
import socket  
import requests  
import json  
from datetime import datetime  
  
target = input("Enter Target IP or Website: ")  
  
common_ports = [21, 22, 23, 25, 53, 80, 110, 135, 139, 143, 443, 445, 8080]  
  
open_ports = []  
  
print("\nScanning Started...\n")  
  
for port in common_ports:  
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)  
    sock.settimeout(1)  
  
    result = sock.connect_ex((target, port))  
  
    if result == 0:  
        print(f"[OPEN] Port {port}")
```

```

        open_ports.append(port)

    sock.close()

headers_info = {}

try:
    response = requests.get(f"http://{target}", timeout=5)

    server = response.headers.get('Server')

    if server:
        headers_info["Server"] = server
        print(f"\nServer Information: {server}")

except:
    print("\nUnable to fetch server headers.")

vulnerabilities = []

if 21 in open_ports:
    vulnerabilities.append("FTP Port Open - Possible insecure configuration")

if 23 in open_ports:
    vulnerabilities.append("Telnet Port Open - Insecure protocol detected")

if 80 in open_ports and 443 not in open_ports:
    vulnerabilities.append("HTTP without HTTPS detected")

report = {
    "Target": target,
    "Scan Time": str(datetime.now()),
    "Open Ports": open_ports,
    "Server Info": headers_info,
    "Vulnerabilities": vulnerabilities
}

with open("report.json", "w") as file:
    json.dump(report, file, indent=4)

print("\nScan Completed!")
print("Report saved as report.json")

```

Security Features

Feature	Description
Open Port Detection	Checks commonly vulnerable ports
Weak Configuration Detection	Detects insecure services like FTP and Telnet
HTTP Security Check	Warns if HTTPS is not enabled
Report Generation	Creates a JSON vulnerability report

Expected Outcome

Gain insights into penetration testing and vulnerability assessment.

Conclusion

The Vulnerability Scanner mini project provides basic knowledge of penetration testing and vulnerability assessment. It helps identify open ports, insecure services, and weak configurations in a simple and educational manner.